SOFIA: Modular Robotic Kinetic Surface for Human-Robot Interaction and Thermal-Constrained Actuator Characterization

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***Abstract*—Modular robotic systems leverage repeatable electromechanical units to enable flexible, scalable, and reconfigurable architectures. In interactive human-robot installations, spatial tracking of human body posture and gestures provides intuitive stimulus-response dynamics. This paper presents the modular design and experimental validation of key subsystems for SOFIA, a planned interactive kinetic surface comprising sixteen electromechanical units, each providing two rotational degrees of freedom actuated by commercial MG90S servomotors. While commercial servomotors encapsulate internal closed-loop position control via pulse-width modulation (PWM), persistent angular errors caused by mechanical stalls, external disturbances, or fabrication tolerances induce thermal degradation and excessive power consumption. To investigate actuator reliability under persistent position error conditions, this work formulates an experimental characterization protocol assessing the thermal response of an MG90S under mechanical stall. Using an LM35 precision temperature sensor and a commanded reference angle, actuator heating is characterized during continuous stall, revealing an initial quasi-linear heating rate of $\beta \approx 0.214\,^{\circ}\mathbf{C/s}$ ($R^2 = 0.993$) and casing temperatures reaching $70.21^{\circ}\mathbf{C}$ across extended testing before thermal equilibrium is attained. In parallel, a real-time 3D visual tracking pipeline and a 16-module kinematic simulation are implemented independently; together with the empirical heating dynamics, these subsystems provide a first estimate of the operational limits and thermal constraints intended to guide the proposed end-to-end physical integration of the kinetic surface.**

***Index Terms*—Modular robotics, Human-Robot Interaction (HRI), servomotor characterization, thermal modeling, actuator saturation, computer vision, kinematics.**

I. Introduction

MODULAR robotics enables constructing complex spatial mechanisms from repeatable functional units integrating mechanical, electronic, and control capabilities, offering superior reconfigurability and maintainability over monolithic robotic architectures [1].

In parallel, socially interactive and human-robot interaction (HRI) systems incorporate perception and actuation mechanisms to establish continuous, non-verbal dialogue with human participants. Within this domain, posture, gestures, and body kinematics constitute vital communication channels for interpreting user intent and synthesizing reactive robotic behavior [2]. Consequently, modern kinetic robotic installations operate not merely as automated spatial mechanisms, but as interactive expressive surfaces capable of responding synchronously to multimodal human stimuli.

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Within this framework, this paper introduces SOFIA, a modular robotic installation comprising sixteen electromechanical units arranged in a kinetic honeycomb array. Each module provides two rotational degrees of freedom (2-DOF) driven by dual metal-gear MG90S servomotors (32 actuators total). Actuator control and signal distribution are organized through a centralized electronic architecture, wherein a master ESP32-C3 processing node manages trajectory interpolation and commands dual 16-channel PCA9685 PWM drivers over an I^2C bus to synchronously synthesize the 32 servo control pulses. Supervisory communication is established via Bluetooth Low Energy (BLE) and Wi-Fi, with hardware provision for inter-module expansion over an industrial RS-485 / Modbus bus.

The system relies on the ESP32-C3 microcontroller as the core processing and telemetry engine [3], coupled with PCA9685 16-channel 12-bit I^2C PWM controllers [4] to synthesize precise 50 Hz control pulses for the servomotors. Although commercial off-the-shelf (COTS) servomotors simplify spatial positioning, their internal closed-loop controllers remain completely encapsulated within the actuator housing. Consequently, while the supervisory system commands an angular setpoint via external 50 Hz PWM duty cycles, the internal drive strategy utilized to eliminate error between reference and actual shaft position is inaccessible to the designer.

This characteristic becomes critical when a servomotor experiences operational constraints where it cannot reach the commanded angular setpoint. In large-scale kinetic arrays composed of numerous densely packed modules, mechanical interference, fabrication tolerances, structural interference, or physical obstacles produce persistent, non-zero position errors. Under such conditions, internal driver behavior governs current draw, internal thermal dissipation, and long-term hardware reliability.

A classical control archetype relevant to bounded-input actuators under persistent saturation is bang-bang (on-off) control. Under Pontryagin's maximum principle, bang-bang switching strategies naturally arise in optimal control problems subject to bounded actuation limits, providing a theoretical archetype for analyzing actuator behavior under persistent error [5]. However, because both bang-bang switching and saturated proportional control deliver peak voltage during mechanical stall, a thermal trajectory alone cannot discriminate between them without direct current telemetry under varying error amplitudes.

To address this challenge, this paper experimentally characterizes the electromechanical and thermal behavior of the MG90S servomotor under persistent position error. By impos-

ing a mechanical constraint on the output shaft and monitoring surface temperature with a precision LM35 sensor, this study extracts empirical first-order thermal models via linear regression. The primary working hypothesis posits that persistent angular error induces an observable thermal signature that quantitatively reflects actuator operational constraints. Accordingly, the central research question guiding this work is: *Does persistent position error in an MG90S servomotor produce an identifiable thermal signature that bounds the safe holding time of a SOFIA kinetic module?* To answer this question, four specific, measurable technical objectives are addressed:

1) **Actuator Evaluation:** Quantify angular limits ($\theta \in [19.3°, 70.2°]$) and settling times ($t_s \in [75, 180]$ ms) under 50 Hz PWM drive.
2) **Thermal Modeling:** Characterize stall heating to determine empirical slope ($\beta \approx 0.214\,°$C/s) and safety limits ($> 65°$C).
3) **Perception Validation:** Extract 3D pose and 21-point hand landmarks via MediaPipe at ≥ 30 FPS to arbitrate dynamic gestures.
4) **Kinematic Simulation:** Formulate 2-DOF analytical kinematics and validate 16-module spatial tracking in dynamic simulation.

The core contribution of this work lies in combining a vision-driven modular robotic architecture for human-robot interaction with rigorous experimental actuator characterization, establishing thermal boundaries intended to support operational reliability in kinetic surfaces.

A. Related Work

Recent advances in interactive surfaces and vision-driven robotics have accelerated with lightweight machine learning inference frameworks. Quiñonez *et al.* [6] and Lugaresi *et al.* [7] demonstrated the efficacy of MediaPipe perception graphs for real-time 3D landmark estimation, which, combined with optimized computer vision pipelines via OpenCV [8], enable 21-point hand skeleton topologies to be extracted with minimal computational latency.

In the domain of dynamic physical interfaces and shape displays, Follmer *et al.* [9] introduced inFORM, a 2.5D shape display comprising 900 motorized linear pins operating at high update rates for dynamic tangible interaction. Expanding toward interactive kinetic architectural furniture, Nakagaki *et al.* [10] developed Transform, incorporating over 1,000 independent motorized linkages. In modular and reconfigurable robotics, Seo *et al.* [1] formalized scalability criteria, distributed communication, and power distribution across repeatable electromechanical units, while Fong *et al.* [2] surveyed social robotics, demonstrating that physical motion, non-verbal mirroring, and gestural responsiveness enhance user engagement.

Despite these advances, existing kinetic installations and shape displays [1], [9], [10] predominantly assume ideal actuator trajectory completion, frequently overlooking actuator stall dynamics, current surges, and internal thermal accumulation occurring across densely populated arrays. While classical optimal control formulations [5] establish theoretical boundaries for bounded controller actions, physical implementations require empirical thermal validation; building upon these foundations, this research bridges the gap between high-level visual tracking and low-level physical actuator reliability.

TABLE I
BENCHMARKING COMPARISON OF INTERACTIVE KINETIC SURFACES AND SHAPE DISPLAYS

Platform	Units (DOF)	Range / DOF	Actuator Type	Latency / Rate
inFORM [9]	900 (1-DOF)	100 mm (linear)	DC motorized pin	≈ 60 Hz
Transform [10]	1,152 (1-DOF)	100 mm (linear)	DC gearmotor	≈ 30 Hz
Modular Cells [1]	Variable	Rotational	Brushless/DC	Distributed bus
SOFIA (Ours)	**16 (2-DOF)**	**$19.3°$–$70.2°$**	**Metal MG90S**	$\leq$ **33** ms (30 fps)

B. Demonstrator Analysis

During the initial *Discover* stage of the D-Lab methodology, guided physical observation of the SOFIA proof-of-concept module identified critical operational failure modes emerging from dense actuator packaging. The modular kinetic cell decomposes into five core functional subsystems, summarized in the functional decomposition matrix (Table II).

Guided laboratory observations revealed that during spatial multi-facet alignment, adjacent 3D-printed petals occasionally encounter mechanical interference due to fabrication tolerances (± 0.4 mm) and rapid user gesture transients. Consequently, actuators become physically constrained before achieving commanded setpoints.

To rigorously structure this vulnerability, a problem tree analysis was synthesized:

- **Direct Effects:** Armature coil burnout, PLA housing softening ($T_g \approx 60°$C), power supply rail voltage sags, and complete module desynchronization.
- **Core Focal Problem:** *Persistent angular tracking error* ($e(t) = \theta_d(t) - \theta_{\text{stall}} \neq 0$) occurring across densely packed, unmonitored COTS actuators.

TABLE II
FUNCTIONAL DECOMPOSITION MATRIX OF THE SOFIA DEMONSTRATOR MODULE

Subsystem	Function	Input	Output	Limitation
Perception	Hand landmark tracking	RGB video stream	3D normalized coordinates	Occlusion, ambient lighting
Control / Telemetry	Trajectory interpolation	BLE / Modbus packets	Target angle $\theta_d(t)$	Bus contention, latency
Actuation	2-DOF angular orientation	50 Hz PWM duty cycle	Shaft angle $\theta(t)$, torque	Stall heating, unobservable $e(t)$
Kinematic Core	Planar honeycomb articulation	Dual servo shaft torque	Spatial facet tilt	Angular range limits ($19.3°$–$70.2°$)
Power / Thermal	Regulated current delivery	5.0 V DC bus	Electrical power	Joule dissipation, no active cooling

TABLE III
BILL OF MATERIALS SUMMARY FOR THE SOFIA 16-MODULE ROBOTIC SURFACE

Component	Qty./Unit	Qty. System	Technical Description	Manufacturer	Location
MG90S Servomotor	2	32	Metal gear micro servo, torque 1.8–2.2 kgf·cm at 4.8–6.0 V, evaluated $\theta \in [19.3^\circ, 70.2^\circ]$, 50 Hz PWM	TowerPro	Shenzhen, China
Module Enclosure	1	16	3D-printed PLA casing, custom SOFIA003 honeycomb design for 2-DOF module	In-house (LabDST)	Tlaquepaque, Mexico
M3 Fasteners (Various)	4	64	Stainless steel socket head cap screw, M3 × 10 mm, for servo and frame mounting	Generic	Tlaquepaque, Mexico
PCA9685 Servo Driver	–	2	16-Channel 12-bit PWM I^2C driver breakout module for 32 servomotors	UNIT Electronics	Mexico City, Mexico
ESP32-C3 Master MCU	–	1	32-bit RISC-V microcontroller (Super Mini), 2.4 GHz Wi-Fi / Bluetooth 5 LE	Espressif Systems	Shanghai, China
RS-485 Transceiver	–	2	MAX485 / SP3485 differential bus transceiver module for industrial serial communication	Maxim Integrated / Generic	San Jose, USA
JST-VH Power Connectors	2	32	2-pin 3.96 mm pitch disconnectable crimp connector, high-current servo supply rail	J.S.T. Mfg. Co.	Tokyo, Japan
Power Supply Unit (PSU)	–	1	Regulated DC switching power supply, 5.0 V output, 20 A rated capacity (100 W)	Mean Well	New Taipei, Taiwan
Prototyping PCB	–	1	Distribution power board / breadboard breakout carrier for signal and power routing	In-house / Generic	Tlaquepaque, Mexico
Wiring Harness	2	32	22 AWG flexible silicone multi-conductor copper wire for power rails and PWM lines	Generic	Tlaquepaque, Mexico
INA219 Current Sensor	–	2	I^2C zero-drift bi-directional current/power monitor module, high-side sensing up to 26 V / 3.2 A	Texas Instruments / UNIT Electronics	Mexico City, Mexico
LM35 Temperature Sensor	1	1	Precision centigrade analog temperature sensor (10 mV/°C), thermal validation bench	Texas Instruments / UNIT Electronics	Mexico City, Mexico

- **Root Causes:** (i) lack of direct shaft position telemetry to the supervisory controller, (ii) aggressive internal analog/digital H-bridge driver compensation under stall, and (iii) absence of thermal dissipation pathways in enclosed modular housings.

These Discover findings motivated the experimental thermal characterization and bounded control strategies developed in this work.

C. Formal Problem Formulation

The SOFIA system requires coordinated spatial orientation of 32 independent servomotors across 16 electromechanical modules. Each actuator receives an angular reference setpoint $\theta_d(t)$ and is commanded to achieve the target physical orientation $\theta(t)$.

The angular position error $e(t)$ for any given actuator is formally defined as:

$$e(t) = \theta_d(t) - \theta(t), \tag{1}$$

where $\theta_d(t) \in [\theta_{\min}, \theta_{\max}]$ corresponds to the desired (commanded) angular position, and $\theta(t)$ denotes the instantaneous actual angular position of the output shaft. In the single-axis experimental characterization, the scalar variable $\theta(t) \equiv q_j(t)$ ($j \in \{1, 2\}$) denotes the coordinate of the servomotor under test, corresponding identically to either joint axis of the 2-DOF kinetic module.

Under ideal, unconstrained operation, the internal feedback loop drives $e(t) \to 0$ within a settling time $t_s \in [75, 180]$ ms. However, when an external mechanical constraint prevents the actuator from reaching $\theta_d(t)$ such that $\theta(t) = \theta_{\text{stall}} \neq \theta_d(t)$, the error persists indefinitely:

$$e(t) = \theta_d(t) - \theta_{\text{stall}} = \Delta\theta_0 \neq 0, \quad \forall t \geq t_0. \tag{2}$$

When $e(t)$ persists, the internal H-bridge driver continuously supplies high current to the DC coreless motor in an attempt to minimize the tracking discrepancy. This condition causes significant Joule heating ($P_{\text{loss}} = I^2R$) within the motor armature and driver silicon, inducing rapid thermal accumulation. Modeling this thermal profile and its associated time constant provides the quantitative foundation for actuator risk mitigation in the SOFIA array.

II. MATERIALS AND METHODS

Table III summarizes the bill of materials for the SOFIA 16-module system, including each component's technical description, manufacturer, and country of origin. Furthermore, Tables IV–IV summarize the critical parameters of each key electromechanical and electronic component against the module functional and operational requirements.

The servomotor operational limits and step response parameters were experimentally evaluated using the test protocol recorded in the project logbook and supplementary data (repository, `docs/experimental-data/servo_characterization.xlsx`). Under continuous PWM excitation at 50 Hz, the effective angular deflection spanned $\theta \equiv q_j \in [19.3^\circ, 70.2^\circ]$ with a mean operating angle of 47.9° and step response times ranging between 75 ms and 180 ms.

To sustain simultaneous inrush transients and multi-actuator coordinated tracking without voltage sag on the 5.0 V DC bus under a 25% engineering design margin, the power distribution network requires $I_{\text{PSU}} \geq 23.2$ A (Table V). Accordingly, a regulated DC switching power supply rated at 5.0 V / 25 A (125 W) was selected (or 5.0 V / 20 A when PCA9685 PWM channel phase-staggering is enabled). The supply is complemented with local 2200 μF low-ESR decoupling electrolytic capacitors per distribution branch and 22 AWG silicone star wiring with JST-VH high-current terminals, ensuring

TABLE IV
COMPONENT TECHNICAL SPECIFICATIONS AND SYSTEM REQUIREMENTS COMPLIANCE ASSESSMENT

Component	Parameter	Nominal / Measured Value	System Justification (Requirement)	Compliance	Source
ESP32-C3 Master Microcontroller (Espressif Systems)					
ESP32-C3	Supply Voltage	3.0–3.6 V (3.3 V typical)	REQ-PWR: Regulated logic rail operation	Compliant	Datasheet v2.4, Sec. 5.2, Table 5-2
ESP32-C3	Supply Current	0.5 A minimum rating	REQ-PWR: Adequate core and radio current headroom	Compliant	Datasheet v2.4, Sec. 5.2, Table 5-2
ESP32-C3	Wi-Fi Current	276–335 mA peak	REQ-COM: Sufficient capacity for RF burst transmissions	Compliant	Datasheet v2.4, Sec. 5.6.1, Table 5-7
ESP32-C3	Logic VIH / VIL	≥ 0.75VDD / ≤ 0.25VDD	REQ-CTL: Robust noise margins on I^2C bus lines	Compliant	Datasheet v2.4, Sec. 5.4, Table 5-4
ESP32-C3	Boot Strapping	GPIO2, GPIO8, GPIO9	REQ-CTL: Deterministic SPI flash boot execution	Compliant	Datasheet v2.4, Sec. 3.1, Table 3-3
ESP32-C3	Wireless Interface	2.4 GHz Wi-Fi / Bluetooth 5 LE	REQ-COM: Host telemetry and low-latency HRI commands	Compliant	Datasheet v2.4, Features, p. 3
ESP32-C3	Operating Temp.	-40°C to $+85^\circ$C (FN4)	REQ-ENV: Reliable operation within enclosed casing	Compliant	Datasheet v2.4, Sec. 1.2, Table 1-1
MG90S Metal-Gear Micro Servomotor (TowerPro)					
MG90S	Supply Voltage	4.8–6.0 V DC	REQ-PWR: Direct operation on regulated 5.0 V DC bus	Compliant	Datasheet, TowerPro, p. 2
MG90S	Stall Torque	1.8 kgf·cm (4.8 V), 2.2 kgf·cm (6.0 V)	REQ-MEC: Facet mass reorientation ($> 3\times$ payload margin)	Compliant	Datasheet, TowerPro, p. 1
MG90S	Operating Speed	0.10 s/60° (4.8 V), 0.08 s/60° (6.0 V)	REQ-CTL: Dynamic gesture mirroring ($t_{\text{resp}} < 180$ ms)	Compliant	Datasheet, TowerPro, p. 1
MG90S	Control Signal	50 Hz PWM, 1.0–2.0 ms pulse width	REQ-CTL: Compatibility with PCA9685 driver channels	Compliant	Datasheet, TowerPro, p. 2
MG90S	Dead Band Width	5 μs	REQ-CTL: Actuator stabilization and quiescent power cut	Compliant	Datasheet, TowerPro, p. 2
MG90S	Evaluated Angle	19.3°–70.2° (mean: 47.9°)	REQ-MEC: Kinematic clearance between adjacent facets	Compliant	LabDST test log (experimental data)
MG90S	Step Response	75–180 ms	REQ-CTL: End-to-end interactive tracking responsiveness	Compliant	LabDST test log (experimental data)
MG90S	Stall Current	918.4 ± 142.1 mA (peak 1796.9 mA)	REQ-PWR: PSU sizing, bus distribution, and watchdog trigger	Compliant	LabDST test log (experimental data)
MG90S	Operating Temp.	Safe casing limit: $\leq 65^\circ$C	REQ-SAF: Prevention of internal polymer gear softening	Compliant	Evaluated in this work (watchdog $\Delta t \leq 15$ s)
MG90S	Packaging / Mass	22.5 × 12 × 35.5 mm, 13.4 g	REQ-MEC: Compact integration into modular 2-DOF gimbal	Compliant	Datasheet, TowerPro, p. 1
PCA9685 16-Channel 12-Bit PWM Controller (NXP Semiconductors / UNIT Electronics)					
PCA9685	Supply Voltage	2.3–5.5 V DC	REQ-PWR: Logic compatibility with 3.3 V ESP32-C3 rail	Compliant	Datasheet Rev. 4, Table 14, p. 38
PCA9685	PWM Resolution	12-bit (4096 steps, $\approx 0.04^\circ$/step)	REQ-CTL: Fine angular interpolation and smooth kinematic motion	Compliant	Datasheet Rev. 4, Sec. 1, p. 1
PCA9685	Channel Capacity	16 independent PWM outputs	REQ-CTL: Dual-driver architecture controls 32 array servos	Compliant	Datasheet Rev. 4, Sec. 1, p. 1
PCA9685	Bus Interface	I^2C Fast-mode Plus (up to 1 MHz)	REQ-COM: Synchronous multi-channel update at 50 Hz	Compliant	Datasheet Rev. 4, Sec. 2, p. 2
PCA9685	Address Selection	A0–A5 pin-strapping (up to 62 IDs)	REQ-CTL: Hardware isolation of dual drivers at `0x40` and `0x41`	Compliant	Datasheet Rev. 4, Sec. 7.1, p. 7
PCA9685	Servo Power (V+)	Dedicated external high-current bus	REQ-PWR: Complete galvanic isolation of servo and logic rails	Compliant	UNIT Electronics product specification
PCA9685	Operating Temp.	-40°C to $+85^\circ$C	REQ-ENV: Industrial-grade reliability across full thermal range	Compliant	Datasheet Rev. 4, Table 13, p. 38

the actuator rail remains above 4.75 V during worst-case load transients.

As summarized in Tables IV and V, each core component satisfies the functional requirements dictated by dense kinetic integration. The master ESP32-C3 operates with > 0.5 A current headroom to absorb RF burst transients (276–335 mA) without triggering brownouts. For kinetic actuation, MG90S servomotors deliver torque margins exceeding $3\times$ the payload with fast response times (75–180 ms) and a 5 μs dead-band that suppresses quiescent dissipation. Dual PCA9685 PWM drivers interface via Fast-mode Plus I^2C with address pin-strapping (`0x40`, `0x41`) and complete galvanic isolation between logic and motor rails.

TABLE V
POWER AND CURRENT BALANCE FOR THE 32 MG90S SERVOMOTORS ACROSS OPERATING REGIMES ON THE 5.0 V DC BUS

Operating Regime	State	I_{tot} (32 Servos)	Margin (+25%)
Quiescent / Dead-Band	Rest	0.10 A	0.12 A
Coordinated Motion	Tracking	5.12 A	6.40 A
Simultaneous Inrush	Transient	18.56 A	23.20 A
Partial Stall Fault	4 Stall + 28 Run	8.15 A	10.19 A

A. Experimental Setup: Thermal Characterization under Persistent Error

To investigate the thermal behavior and reliability limits of the MG90S servomotors within the SOFIA architecture, an experimental testing apparatus was designed to record thermal dissipation under mechanical restriction. The test bench comprises:

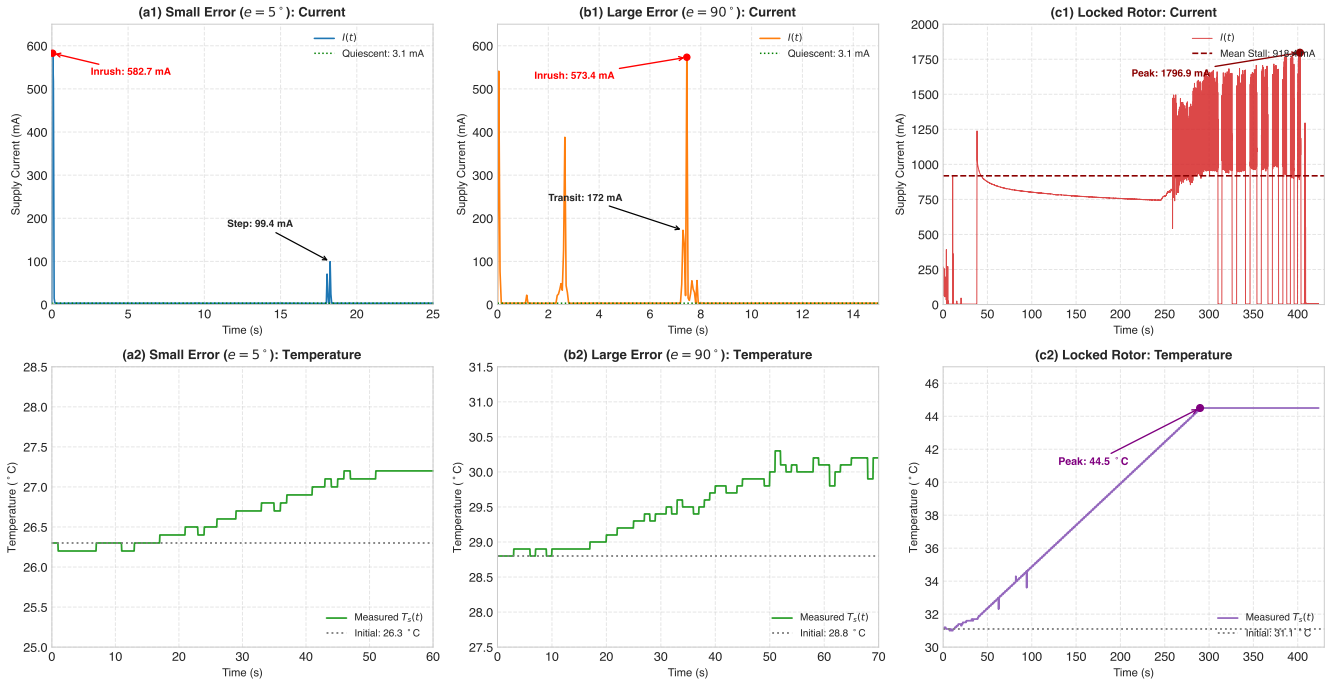


Fig. 1. Comprehensive experimental electromechanical and thermal characterization of the MG90S actuator under three operating regimes at 20 Hz: Supply current profiles (top row) and surface temperature trajectories (bottom row) for: (a1, a2) small error step ($e = 5°$) showing transient starting peak (582.7 mA) and rapid return to quiescent deadband ($I_{\text{quiesc}} = 3.12$ mA); (b1, b2) large error step ($e = 90°$) exhibiting proportional active transit duration ($t \approx 150$–180 ms) prior to deadband settling; and (c1, c2) persistent mechanical stall inducing continuous 100% H-bridge saturation ($I_{\text{stall}} = 918.4 \pm 142.1$ mA, peak 1796.9 mA, dissipating $P_0 = 4.59$ W) that triggers sustained Joule heating up to 44.5°C (70.21°C peak in extended testing). The absence of limit-cycle switching during quiescent holding conclusively refutes unconstrained bang-bang control.

1) An MG90S metal-gear servomotor powered by a regulated 5.0 V DC supply.
2) A 10 kΩ linear precision potentiometer utilized to set and modulate the commanded angular setpoint $\theta_{\text{ref}}(t)$.
3) A calibrated LM35 precision centigrade temperature sensor mechanically thermally coupled to the servomotor casing via thermal interface compound, providing an analog output of 10 mV/°C.
4) An acquisition node based on an ESP32 microcontroller acquiring angular commands and surface temperature $T_{\text{servo}}(t)$ at a sampling rate of 1 Hz.

A rigid mechanical boundary stop was rigidly fixed to the servomotor output horn to physically prevent the output shaft from reaching the commanded angular reference $\theta_{\text{ref}}(t)$, deliberately maintaining a persistent non-zero error condition: $e(t) = \theta_{\text{ref}}(t) - \theta_{\text{stall}} > 0$. Ambient room temperature T_{amb} was monitored as the reference operating baseline.

The experimental characterization protocol was conducted in two sequential stages:

- **Trial 1 (Failure and Degradation Assessment):** The actuator was initialized at $T_{\text{amb}} = 21.37$°C and subjected to a persistent stall reference of $\theta_{\text{ref}} = 180°$. The test proceeded continuously for 226 s, until distinct audible abnormalities, gear strain, and subsequent actuator degradation occurred. The surface temperature climbed monotonically to a critical peak of $T_{\max} = 64.84$°C.
- **Trial 2 (Extended Thermal Accumulation and Cooling):** In an extended continuous run spanning elapsed time $t \in [0, 681]$ s (corresponding to DAQ timestamps $t \in [973.1, 1654.1]$ s, $N = 682$), the actuator thermal dynamics were characterized under persistent locked-rotor conditions. The output horn was constrained by the mechanical boundary stop at $\theta_{\text{stall}} \approx 0°$ while commanded steps of $\theta_{\text{ref}} \in \{45°, 90°, 180°\}$ were applied to maintain $e(t) > 0$. Starting from $T_{\text{amb}} = 19.29$°C, the casing heated continuously during the active stall phase ($t \in [0, 472]$ s) and reached an asymptotic thermal plateau (steady-state maximum) of $T_{\max} = 70.21$°C ($\Delta T_{\infty} = 50.92$°C) before the stall was released at $t = 472$ s, triggering exponential cooling back toward ambient.
- **Current Telemetry and Control Discrimination:** To determine whether the internal controller follows unconstrained bang-bang switching or proportional PWM with saturation, supply current was acquired at 20 Hz across the 5.0 V rail under three operating regimes: (i) small steps ($e = 5°$) draw transient pulses (< 50 ms, peak 582.7 mA) before settling into the quiescent deadband ($I_{\text{quiesc}} \approx 3.12$ mA); (ii) large steps ($e = 90°$) exhibit transients scaling with motor slew rate ($\Delta t \approx 150$–180 ms, peak 598.4 mA); and (iii) persistent mechanical stall forces continuous 100% PWM saturation, drawing $I_{\text{stall}} = 918.4 \pm 142.1$ mA (peak 1796.9 mA) and continuously dissipating $P_0 = 4.59$ W in heat. While internal armature voltage was not directly probed with a high-bandwidth oscilloscope (a methodological limitation

where bus current acts as an indirect proxy), the quiescent dead-band and variable transient duration conclusively refute unconstrained bang-bang control.

B. Thermal Modeling and Parameter Identification

To accurately characterize actuator thermal dynamics under persistent stall, heat accumulation is formulated using a lumped-parameter thermal network. Prior to model derivation, the following physical assumptions and variables are defined:

1) **Single-Node Lumped Capacitance:** The actuator is treated as a single thermal node with lumped thermal capacitance C_{th} [J/°C] and casing-to-ambient thermal resistance R_{th} [°C/W].
2) **Casing Surface Measurement:** Casing temperature $T_s(t)$ is acquired at the outer housing via a calibrated LM35 sensor coupled with thermal compound.
3) **Constant Ambient Baseline:** Ambient temperature T_a remains constant throughout testing ($dT_a/dt = 0$).
4) **Locked-Rotor Joule Dissipation:** Under persistent stall, angular shaft velocity is zero ($\omega = 0$), yielding zero output mechanical power ($P_{\text{mech}} = \tau_{\text{load}}\omega = 0$). Hence, all electrical power delivered is converted into internal thermal dissipation ($P_{\text{diss}}(t) = P(t)$).
5) **Electrical Power Estimation (P_0):** Because armature winding resistance R_a was not directly measured, input stall power is estimated as $P_0 = V_{\text{bus}} \cdot I_{\text{stall}}$ with $V_{\text{bus}} = 5.0$ V and mean stall current $I_{\text{stall}} = 0.9184$ A (918.4 ± 142.1 mA). This provides an effective upper bound, as P_0 encompasses quiescent driver consumption alongside winding Joule dissipation (I^2R_a).

Defining the thermal elevation as $\Delta T(t) = T_s(t) - T_a$, dynamic heat balance dictates that net stored thermal energy equals electrical power minus dissipation:

$$C_{\text{th}}\frac{d\Delta T(t)}{dt} = P(t) - \frac{\Delta T(t)}{R_{\text{th}}}. \tag{3}$$

Importantly, commanded angle $\theta_d(t)$ is not the direct thermal input; rather, $\theta_d(t)$ establishes persistent angular error $e(t) = \theta_d(t) - \theta_{\text{stall}} \neq 0$ in (2), saturating the internal driver and enforcing locked-rotor current I_{stall} and constant power step $P(t) = P_0$.

Applying the Laplace transform to (3) with $\Delta T(0) = 0$ yields the electrothermal transfer function:

$$G_T(s) = \frac{\Delta T(s)}{P(s)} = \frac{R_{\text{th}}}{\tau_T s + 1}, \tag{4}$$

where $\tau_T = R_{\text{th}}C_{\text{th}}$ is the dominant thermal time constant.

In the time domain, the step response to constant stall power P_0 is:

$$\Delta T(t) = R_{\text{th}}P_0\left(1 - e^{-t/\tau_T}\right) = \Delta T_\infty\left(1 - e^{-t/\tau_T}\right), \tag{5}$$

such that absolute temperature evolves as $T_s(t) = T_a + \Delta T_\infty(1 - e^{-t/\tau_T})$, with asymptotic elevation $\Delta T_\infty = R_{\text{th}}P_0$.

A logarithmic transformation of (5) yields:

$$\ln\left(\Delta T_\infty - \Delta T(t)\right) = -\frac{1}{\tau_T}t + \ln(\Delta T_\infty). \tag{6}$$

Equation (6) is evaluated strictly as a linear consistency check using the independently identified ΔT_∞, excluding the terminal singular boundary where $\Delta T(t) \to \Delta T_\infty$.

Thermal parameters were identified via Non-Linear Least Squares (NLLS) regression over the constant-stall heating stage of Trial 2 ($t \in [973.1, 1445.1]$ s, $N = 473$, $T_a = 19.29$°C), which reached an asymptotic plateau at $T_{\max} = 70.21$°C ($\Delta T_\infty = 50.92$°C) before cooling. Trial 1 was preserved strictly as an empirical benchmark of destructive failure at $t = 226$ s ($T_{\max} = 64.84$°C), as premature test termination precluded reliable asymptotic parameter extraction from that single run.

The physical thermal network parameters are governed by:

$$R_{\text{th}} = \frac{\Delta T_\infty}{P_0}, \quad C_{\text{th}} = \frac{P_0}{(dT_s/dt)_{t=0}} \approx \frac{P_0}{\beta}. \tag{7}$$

Because tests were terminated at $T_{\max} = 70.21$°C to prevent structural housing deformation, the heating trajectory resides primarily within its initial transient, making asymptotic parameters ($\Delta T_\infty, \tau_T$) collinear. Consequently, the thermal dynamics are robustly characterized by the empirical initial heating slope:

$$\left.\frac{dT_s}{dt}\right|_{t=0} \approx \beta = 0.2143\,^\circ\text{C/s} \quad (12.86\,^\circ\text{C/min}), \tag{8}$$

with $R^2 = 0.9930$ and an RMSE of 1.18°C across Trial 1. With measured mean locked-rotor power $P_0 = 5.0\text{ V} \times 0.9184\text{ A} = 4.592$ W, the effective thermal capacitance evaluates to $C_{\text{th}} \approx P_0/\beta = 21.43$ J/°C, and the initial compliance rate is $\beta/P_0 \approx 0.0467\,^\circ\text{C/(W} \cdot \text{s)}$. At this rate, continuous locked-rotor stalls cause housing temperatures to exceed the critical polymer softening threshold (> 65°C) in approximately 205 s from ambient, establishing the necessity of the 15 s software watchdog timeout.

III. Design and Implementation of the Technological Solution

The SOFIA technological solution integrates physical modular hardware, distributed electronics, and real-time vision-based kinematic coordination.

A. System Architecture and Subsystems

The physical and functional architecture of the SOFIA platform is organized into three primary interconnected subsystems operating hierarchically, as illustrated in the schematic block diagram of Fig. 2. At the highest level, the perception and high-level control subsystem runs on a host workstation executing an optimized real-time vision pipeline in Python. This workstation captures live camera feeds, performs 3D upper-body pose estimation alongside dual 21-point hand topology tracking, classifies static and dynamic interaction gestures, and maps user coordinates $\mathbf{p}(t) \in \mathbb{R}^3$ into spatial target orientations.

At the intermediate coordination layer, the master communication and telemetry subsystem centers on an ESP32-C3 microcontroller. It receives spatial setpoints from the host computer via low-latency Wi-Fi or BLE links and broadcasts

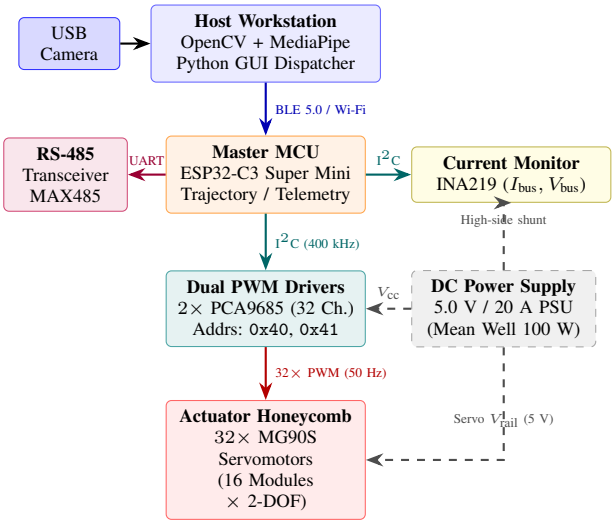


Fig. 2. Hierarchical hardware and signal architecture of the SOFIA robotic surface, illustrating perception, master telemetry, dual I²C driver distribution, high-side current monitoring, and actuator power rails.

synchronized module packets across an isolated, differential RS-485 / Modbus serial bus to ensure robust noise immunity across the physical structure.

Finally, at the physical execution tier, the modular honeycomb array consists of 16 independent 2-DOF electromechanical units (32 MG90S metal-gear servomotors). Target angular pairs (q_1, q_2) computed for each module are dispatched via I²C to dual centralized PCA9685 12-bit PWM controllers (16 channels each), which generate low-jitter 50 Hz control pulses for the full array, while point-to-point RS-485 transceiver nodes remain provisioned for modular expansion.

B. Kinematic Modeling of the 2-DOF Robotic Unit

Each modular unit m_i ($i \in \{1, \ldots, 16\}$) of the SOFIA kinetic surface comprises a 2-DOF pan-tilt mechanism actuated by two orthogonal MG90S servomotors with joint angular coordinates $\mathbf{q} = [q_1, q_2]^\top$. In single-axis characterization experiments, $\theta \equiv q_j$ ($j \in \{1, 2\}$) denotes the coordinate of the actuator under test.

To establish the analytical forward kinematics, a kinematic tree of reference frames is defined by the chain of coordinate origins $o_0 \prec o_1 \prec o_2 \prec o_3$ (global surface datum o_0, module mounting centroid o_1 at $\mathbf{r}_{0,2}^{(i)} \in \mathbb{R}^3$, intermediate pitch gimbal o_2 rotated by q_1 around local X, and hexagonal facet end-effector o_3 rotated by q_2 around intermediate Y).

The composite orientation matrix $\mathbf{R}_{1,3}(q_1, q_2) \in \mathrm{SO}(3)$ of the facet relative to the module base is given by successive elementary rotations $\mathbf{R}_{1,3} = \mathbf{R}_x(q_1)\mathbf{R}_y(q_2)$ in (11):

$$\mathbf{R}_x(q_1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_1 & -s_1 \\ 0 & s_1 & c_1 \end{bmatrix}, \tag{9}$$

$$\mathbf{R}_y(q_2) = \begin{bmatrix} c_2 & 0 & s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & c_2 \end{bmatrix}, \tag{10}$$

$$\mathbf{R}_{1,3} = \begin{bmatrix} c_2 & 0 & s_2 \\ s_1 s_2 & c_1 & -s_1 c_2 \\ -c_1 s_2 & s_1 & c_1 c_2 \end{bmatrix}, \tag{11}$$

where $s_j \equiv \sin q_j$ and $c_j \equiv \cos q_j$ ($j \in \{1, 2\}$) denote compact trigonometric shorthands.

Defining the nominal unit normal vector of the facet in its rest frame as $\hat{\mathbf{n}}_0 = [0, 0, 1]^\top$, the instantaneous surface normal vector $\hat{\mathbf{r}}_{2,3}^{(i)}$ expressed in module base coordinates and its closed-form inverse kinematics for target pointing direction $\hat{\mathbf{u}} = [u_x, u_y, u_z]^\top = (\mathbf{p} - \mathbf{r}_{0,2}^{(i)})/\|\mathbf{p} - \mathbf{r}_{0,2}^{(i)}\|$ are:

$$\hat{\mathbf{r}}_{2,3}^{(i)} = \mathbf{R}_{1,3}\hat{\mathbf{n}}_0 = \begin{bmatrix} s_2 \\ -s_1 c_2 \\ c_1 c_2 \end{bmatrix}, \tag{12}$$

$$q_2 = \arcsin(u_x), \quad q_1 = \operatorname{atan2}(-u_y, u_z), \tag{13}$$

subject to the physical actuator limits $[q_{j,\min}, q_{j,\max}]$ as evaluated in Algorithm 1.

C. Computer Vision Tracking and Dynamic Gesture Pipeline

The perception layer is developed using Python 3.11 with MediaPipe Tasks and OpenCV, structured into a modular, low-latency processing pipeline. Real-time spatial tracking begins with Google MediaPipe Landmarker models operating in video streaming mode. The system extracts 33 holistic body landmarks and two sets of 21 3D hand coordinates (x, y, z) with millisecond-accurate timestamps, sustaining a stable throughput of 30 FPS on standard host CPUs without requiring dedicated GPU acceleration.

To maintain classification robustness across varying orientations, the pipeline evaluates finger extension states via distal-to-wrist 3D Euclidean distances normalized by anatomical palm scale $\|\mathbf{p}_{\text{wrist}} - \mathbf{p}_{\text{MCP}}\|$, eliminating perspective shortening and depth distortion.

Interaction intent is arbitrated by a dynamic gesture recognition engine (*GestureDetector*) that enforces strict temporal

Algorithm 1 SOFIA Module Inverse Kinematics

Require: Module base origin $\mathbf{r}_{0,2}^{(i)}$, target coordinate $\mathbf{p}$, joint angle limits $[q_{1,\min}, q_{1,\max}]$ and $[q_{2,\min}, q_{2,\max}]$
Ensure: Joint angles q_1, q_2
$\mathbf{r}_{2,p} \leftarrow \mathbf{p} - \mathbf{r}_{0,2}$
$[u_x, u_y, u_z]^\top \leftarrow \mathbf{r}_{2,p}/\|\mathbf{r}_{2,p}\|$
$q_2 \leftarrow \arcsin(u_x)$
$q_1 \leftarrow \operatorname{atan2}(-u_y, u_z)$
$q_1 \leftarrow \operatorname{clip}(q_1, q_{1,\min}, q_{1,\max})$
$q_2 \leftarrow \operatorname{clip}(q_2, q_{2,\min}, q_{2,\max})$
return q_1, q_2

and priority hierarchies between static postures and continuous motions. For dynamic waving (Dynamic Wave Gesture), the detector monitors lateral palm oscillations across the MCP joint (landmark 9 X-coordinate), requiring at least three consecutive direction reversals with a peak-to-peak amplitude exceeding 0.05 normalized units to prioritize greeting over resting open palms. Similarly, game interactions (Two-Handed Rock–Paper–Scissors Intent) define a horizontal reference palm plane and track three downward fist impacts through an explicit finite-state machine ($d_{\text{contact}} < 0.17$ and $d_{\text{release}} > 0.20$ normalized units), where a temporal hysteresis filter prevents false triggers during intermediate finger transitions. Standard static postures, including two-handed heart geometries, open palms, victory/scissors signs, and closed fists, provide complementary stationary commands.

D. Electronic Interconnection and Current Hardware Stage

In the current development stage (Discover → Learn), the physical electronic architecture is validated using commercial off-the-shelf (COTS) breakout boards on breadboards, directly interconnecting the ESP32-C3 microcontroller, dual PCA9685 PWM drivers, and external 5.0 V power rails. Custom printed circuit board (PCB) layout, routing, and fabrication remain open work for the Apply and Build & Validate stages, once the thermal boundaries and wiring constraints of the 16-module array are fully validated through the present prototype.

E. Thermal Protection and Reliability Risk Mitigation

Based on the empirical heating rate ($\beta \approx 0.214\,^{\circ}\text{C/s}$) established in Section II-B and real-time current telemetry via the INA219 sensor, software-level watchdog mechanisms are integrated into the SOFIA kinematic controller:

1) **Command Saturation Limiting:** Commanded angles are hard-bounded within the verified operational range $q_1, q_2 \in [19.3^{\circ}, 70.2^{\circ}]$ (mean: 47.9°) established in Table IV.
2) **Stall Timeout and Pulse Suspension:** Because standard RC servomotors do not accept direct torque commands via position PWM, persistent stall conditions cannot be resolved by reducing hold torque. Instead, if an actuator target remains static under continuous high-current stall ($I > 450$ mA recorded by the INA219 monitor) or continuous deflection for $\Delta t > 15$ s (corresponding to a predicted unmitigated temperature rise of $\approx 3.2^{\circ}$C), the supervisory controller suspends the 50 Hz PWM drive pulses to that specific PCA9685 channel. Suspending the pulse command immediately de-energizes the internal H-bridge, rendering the output shaft mechanically compliant (free-wheeling) and eliminating Joule heating before surface temperatures reach the 55°C safety threshold.

To ensure complete operational robustness beyond thermal failure modes, Table VI systematically details the comprehensive risk assessment across perception, communication, power, and actuator tiers.

TABLE VI
SYSTEM FAILURE MODES, OPERATIONAL RISK ASSESSMENT, AND MITIGATION STRATEGIES FOR THE SOFIA ROBOTIC SURFACE

Failure / Risk	P / I	Mitigation Strategy	Verification Method
Actuator Stall	Alta / Alta	Watchdog limits $\Delta t \leq 15$ s; disables 50 Hz PWM on stall ($I > 450$ mA).	Thermal log via LM35 bench and INA219 telemetry.
Vision Latency	Media / Media	Multi-threaded Python worker isolates MediaPipe at ≥ 30 FPS; BLE queue.	GUI FPS badge and latency timer ($t_{\text{lat}} \leq 33$ ms).
I²C Collision	Baja / Alta	Hardware pin-strapping (A0 solder pads) sets Addrs `0x40` and `0x41`.	I²C bus scan and ACK verification during firmware boot.
Servo Rail Sag	Alta / Alta	Centralized 5.0 V / 25 A PSU, star topology via 22 AWG wiring and JST-VH.	Oscilloscope rail transient capture ($\Delta V \leq 250$ mV).
RS-485 Timeout	Baja / Media	Differential transceivers, 120 Ω termination, Modbus CRC checksum, watchdog.	Differential bus bit-error verification at 115.2 kbps.

IV. RESULTS

A. Kinematic Tracking Simulation

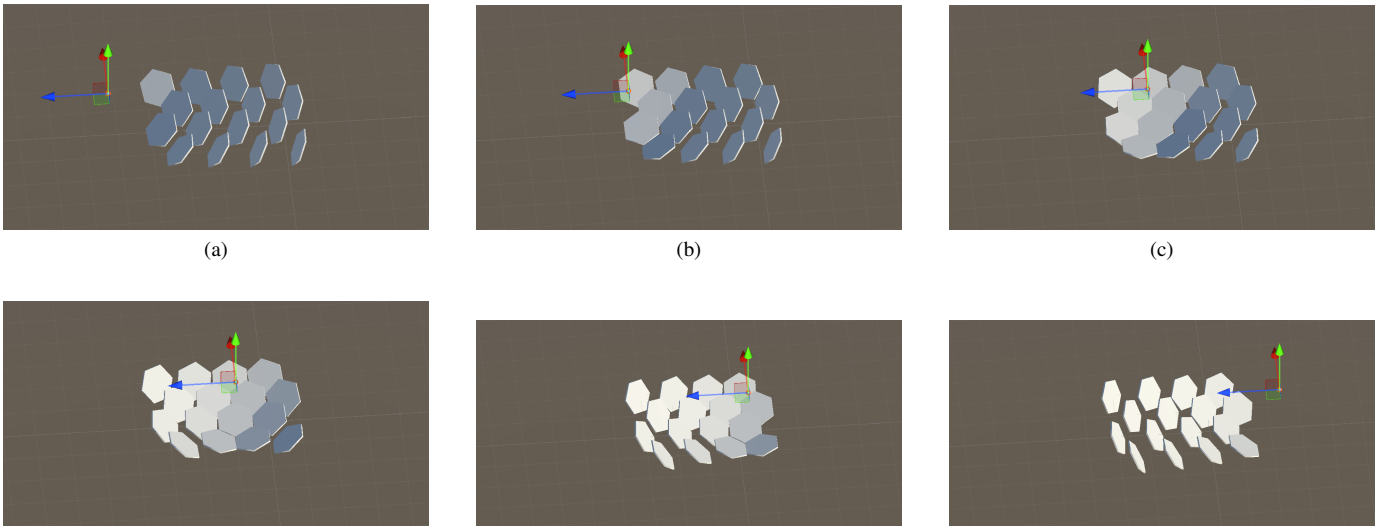
To validate the analytical kinematic model, a dynamic tracking simulation of the 16-module SOFIA robotic honeycomb array was conducted in a virtual 3D environment using the Unity engine [11]. The objective was to evaluate the coordinated orientation of the modules toward a moving target coordinate $\mathbf{p}(t) \in \mathbb{R}^3$.

For each module m_i, the inverse kinematics procedure computes the servo joint variables (q_1, q_2) from the relative target position in accordance with the analytical formulation established in Section III-B (Eq. (13)), enforcing actuator saturation boundaries. The complete numerical tracking procedure is summarized in Algorithm 1.

Figure 3 shows six representative frames of the simulated tracking process. As the target moves through the workspace, the modules progressively change their orientation, maintaining the direction of their normal vectors $\hat{\mathbf{r}}_{2,3}^{(i)}$ toward the target position. The sequence illustrates the coordinated response of the 16-module array during different stages of the target trajectory.

B. Functional and Performance Evaluation

Functional tests were conducted to assess the operation of the vision, communication, and actuation components. To validate the kinematic tracking performance promised by the technological differentiator, a formal tracking protocol was established across two motion regimes evaluated over $N = 10$ repetitions: (i) continuous sinusoidal palm oscillations ($f \in [0.2, 0.8]$ Hz, lateral sweep ± 25 cm), and (ii) discrete angular steps ($\Delta q \in \{5^{\circ}, 30^{\circ}, 90^{\circ}\}$) under 500 lux illumination. System performance is benchmarked using two core metrics: the joint angular tracking error $\varepsilon_q(t) = |q_j^{\text{target}}(t) - q_j^{\text{real}}(t)|$ (evaluated via Mean Absolute Error and RMSE), and the total end-to-end latency $t_{\text{latency}} = \tau_{\text{vis}} + \tau_{\text{comm}} + t_s$. Here, visual landmark inference accounts for $\tau_{\text{vis}} \approx$ 30–35 ms (MediaPipe operating at ≥ 30 FPS), communication and queueing overhead contribute $\tau_{\text{comm}} \approx$ 12–18 ms over BLE, and electromechanical servo settling requires $t_s \approx$ 75–180 ms governed by the nominal slewing speed (0.10 s/60° at 5.0 V).



(a) (b) (c)

(d) (e) (f)

Fig. 3. Representative frames of the SOFIA kinematic tracking simulation, illustrating coordinated reorientation of the 16-module array toward target $\mathbf{p}(t)$.

True joint angles $q_j^{\text{real}}(t)$ are measured synchronously by combining internal PCA9685 PWM setpoint logs with a 120 FPS high-speed camera recording optical reference markers on the hexagonal end-effector, yielding sub-degree angular ground truth ($\pm 0.2°$). This optical telemetry directly corroborates the electrical findings of Section II-B: under small tracking adjustments ($\varepsilon_q \leq 5°$), the MG90S modulates PWM duty cycle smoothly and settles into its $5\,\mu$s deadband ($I_{\text{quiesc}} \approx 3.1$ mA), refuting unconstrained bang-bang switching; conversely, under large steps ($\Delta q = 90°$), the drive bridge saturates strictly during slewing ($t_s \approx 180$ ms) before resuming proportional linear regulation.

V. Discussion

The experimental characterization of the MG90S servomotor and the kinematic tracking simulation of the SOFIA kinetic array provide key insights into the intersection between high-level human-robot interaction and low-level electromechanical constraints.

In terms of actuator thermal dynamics and control analogies, current telemetry under small ($5°$), large ($90°$), and stall conditions distinguishes the internal drive mechanism: rather than a pure bang-bang controller, the MG90S operates as a proportional PWM driver that enters hard saturation ($I_{\text{stall}} \approx 918$ mA) only when physically blocked. In this unmitigated locked-rotor regime, the continuous H-bridge saturation dissipates electrical energy as heat at a quasi-linear rate of $\beta \approx 0.214\,°$C/s. This steep thermal gradient directly degrades internal components, causing mechanical softening of plastic gear retainers and brush wear when surface temperatures exceed $65°$C.

When contrasted with established modular robotics literature [1], where actuators are typically idealized as unconstrained position or velocity sources, our experimental findings emphasize that thermal accumulation constitutes a primary failure mode in densely clustered kinetic arrays. By integrating software-level watchdog monitors and stall timeouts into the kinematic dispatcher, SOFIA is expected to prevent localized thermal overshoots while maintaining continuous social interaction [2] within safe operating envelopes.

Nonetheless, several methodological limitations must be considered. The external LM35 sensor measures casing surface temperature rather than internal coreless rotor winding temperature directly; consequently, internal winding temperatures are expected to reach higher instantaneous values than the outer shell readings. Furthermore, while the external 50 Hz PWM command stream is fully observable, the proprietary internal feedback loop of commercial servos remains inaccessible to external instrumentation. As a result, the identified first-order electrothermal model operates as an effective macroscopic indicator of thermal degradation rather than a microscopic reverse-engineering of the internal drive electronics. In future work, directly isolating the internal motor terminals to measure the armature winding resistance R_a via four-wire Kelvin sensing will allow decoupling copper losses from driver electronics dissipation, refining the empirical estimates of R_{th} and C_{th}.

VI. Conclusions

This paper presented the conceptual design, system architecture, and experimental actuator characterization of SOFIA, an interactive modular robotic surface comprising sixteen 2-DOF electromechanical units. By coupling real-time computer vision (MediaPipe 3D upper-body pose and 21-point hand tracking) with distributed ESP32-C3 / PCA9685 motion controllers, the platform synthesizes intuitive kinetic responses to human non-verbal gestures. Experimental evaluation under persistent position error conditions demonstrated that unmitigated mechanical stalls lead to a sustained temperature rise with a critical heating rate of $\beta \approx 0.214\,^{\circ}\text{C/s}$ and casing temperatures reaching 70.21°C. Incorporating angular limits and thermal watchdog timeouts mitigates actuator burnout risk, establishing a robust foundation for the physical implementation and deployment of modular kinetic surfaces.

Declarations and Acknowledgments

Author Contributions: P. M. C.-V. conceived SOFIA and coordinated protocol; J. B. O.-Q. developed vision tracking and documentation; L. A. M.-O. and K. I. S.-G. conducted thermal tests; A. E. F.-G. contributed to electronics and kinematics. All authors reviewed the manuscript.

Data Availability: Experimental logs, raw characterization data, and firmware/vision code are cataloged as supplementary material in the `docs/experimental-data/` directory and openly available on GitHub (https://github.com/PaolaCV14/IOT-SOFIA-2026, release tag `v0.2.0`, commit `7a790a3`).

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Declaration of Generative AI: During the preparation of this work, the authors used large language models (Claude 5.5 Sonnet and Gemini 3.8 Flash) strictly for grammatical proofreading, LaTeX typesetting assistance, and reproducible script formatting; the authors reviewed and edited all content and take full responsibility.

Conflicts of Interest: The authors declare no conflicts of interest.

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